YAWEN OUYANG

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Research Area: LLM for Scientific Discovery

EXPERIENCE

Researcher — Shanghai AI Lab, AI4S 2025 – Now
Postdoctoral Researcher — Tsinghua University, AI4S 2023 – 2025
Research Intern — ByteDance AI Lab, NLP 2021 – 2022
Ph.D. in Computer Science — Nanjing University, NLP 2018 – 2023
B.Eng. in Software Engineering — Dalian University of Technology 2014 – 2018

CAPABILITIES

- **Protein agent**: developed a protein modification benchmark and built a Protein env for post-training a 300B-parameter model to improve its capabilities [12]. Also contributed to a general-purpose protein foundation model [11].
- **Generative modeling over scientific data**: developed a family of generative models spanning crystals [7], MOFs [10, 8], and molecules [6], and built MoMa [9] as a materials representation model adapted for catalyst discovery [5].
- **OOD detection**: developed OOD detection methods for single-label [2], multi-label [3], unlabeled settings [4].

RESEARCH

Selected papers with non-trivial contributions. *: equal contributions, †: corresponding author, ‡: (co-)mentored student

[12] PFArena: Benchmarking Language Models for Protein Modification. ***Technical Report 2026.***
Yawen Ouyang*, Xinbo Zhang*, Ziyuan Ma*, Yixin Wu, Wenbin Liao, Feiran Zhang, Wenjie Li, et al.

[11] AMix-2: Establishing Protein as a Native Modality in Large Language Models. ***Technical Report 2026.***
Keyue Qiu, Yixin Wu, Lihao Wang, Yawen Ouyang, Jixiang Yu, Zihan Zhou, Changze Lv, Dongyu Xue, et al.

[10] Enhancing Spatial Reasoning in Large Language Models for MOF Structure Prediction. ***KDD 2026.***
Mianzhi Pan‡, JianFei Li, Peishuo Liu, Botian Wang, Yawen Ouyang, Yiming Rong, Hao Zhou, Jianbing Zhang.

[9] MoMa: A Modular Deep Learning Framework for Material Property Prediction. ***ICLR 2026.***
Yawen Ouyang*, Botian Wang*, Yaohui Li*, Yiqun Wang, Haorui Cui, Jianbing Zhang, Xiaonan Wang, et al.

[8] MOF-BFN: Metal-Organic Frameworks Structure Prediction via Bayesian Flow Networks. ***NeurIPS 2025.***
Rui Jiao, Hanlin Wu, Wenbing Huang, Yuxuan Song, Yawen Ouyang, Yu Rong, Tingyang Xu, Pengju Wang, et al.

[7] A Periodic Bayesian Flow for Material Generation. ***ICLR 2025.***
Hanlin Wu‡, Yuxuan Song, Jingjing Gong, Ziyao Cao, Yawen Ouyang†, Jianbing Zhang, Hao Zhou, et al.

[6] Generative AI-Powered Inverse Design for Tailored Narrowband Molecular Emitters. ***Digital Discovery.***
Yawen Ouyang*, Mianzhi Pan*, Tianhao Tan*, Qian Jin, Yougang Chu, Wei-Ying Ma, Jianbing Zhang, et al.

[5] Adaptive and scalable catalyst discovery with composable intelligence. ***Science. (Under review)***
Wentao Li, et al., Yawen Ouyang, Manu Suvarna, Wei-Ying Ma, Ya-Qin Zhang, Kostya Novoselov, et al.

[4] On Prefix-tuning for Lightweight Out-of-distribution Detection. ***ACL 2023.***
Yawen Ouyang, Yongchang Cao, Yuan Gao, Zhen Wu, Jianbing Zhang, Xinyu Dai.

[3] Towards Multi-label Unknown Intent Detection. ***COLING 2022.***
Yawen Ouyang, Zhen Wu, Xinyu Dai, Shujian Huang, Jiajun Chen.

[2] Energy-based Unknown Intent Detection with Data Manipulation. ***ACL 2021*** *Findings.*
Yawen Ouyang, Jiasheng Ye, Yu Chen, Xinyu Dai, Shujian Huang, Jiajun Chen.

[1] Dialogue State Tracking with Explicit Slot Connection Modeling. ***ACL 2020.***
Yawen Ouyang*, Moxin Chen*, Xinyu Dai, Yinggong Zhao, Shujian Huang, Jiajun Chen.

AWARDS & FUNDS

- National Natural Science Foundation of China Youth Program, 2025
- National Scholarship, 2016